

Design and Implementation of a Low-Cost, Bi-Directional Free-Space Optical (FSO) Communication System Using Microcontroller-Modulated Direct Diode Lasers

Ali Shah

Department of Computer Science & Engineering / Applied Physics

September 2026

ABSTRACT

Free-Space Optical (FSO) communication offers a high-bandwidth, line-of-sight alternative to traditional Radio Frequency (RF) links, mitigating spectrum congestion, latency, and susceptibility to electromagnetic interference (EMI). However, commercial FSO hardware remains economically prohibitive for localized, tactical, or educational deployment. This paper presents **Project L.O.K.I.** (Light-Oriented Knowledge Interface), a prototype bi-directional FSO transceiver system constructed using dual ESP32 microcontrollers and direct-diode semiconductor lasers. By employing direct intensity modulation (OOK) coupled with photodiode receiver circuits, the system establishes a full-duplex optical link over line-of-sight distances. We detail the physical-layer hardware architecture, optical signal conditioning, framing protocols, and performance metrics, evaluating throughput, Bit Error Rate (BER), and alignment tolerance. The resulting system demonstrates a robust proof-of-concept for ultra-low-cost, secure, RF-denied optical communication.

Keywords: Free-Space Optics (FSO), Direct Modulation, ESP32, Photodiode Transimpedance Receiver, Intensity Modulation/Direct Detection (IM/DD), Optical Alignment.

1. INTRODUCTION

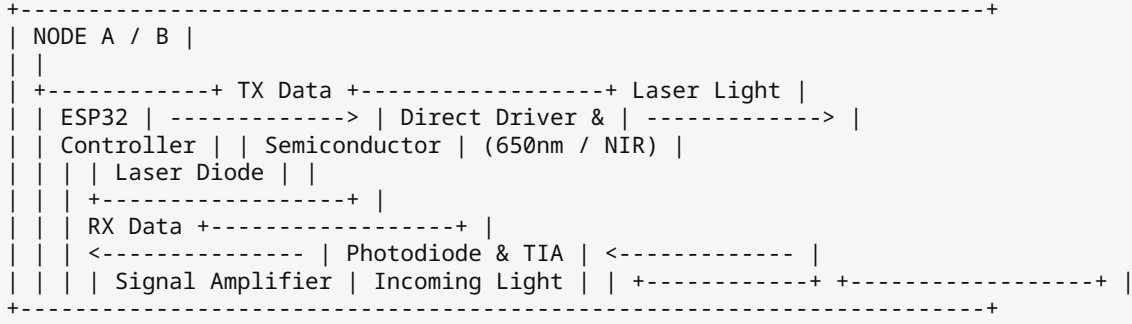
With the exponential growth of connected nodes and wireless data traffic, the Radio Frequency (RF) spectrum faces severe saturation. In environments characterized by heavy RF interference, strict security requirements, or physical containment (such as Faraday-shielded facilities or industrial floors), RF communication degrades significantly.

Free-Space Optics (FSO) leverages visible or near-infrared (NIR) light to transmit data through unguided atmospheric channels. FSO benefits from an unlicensed, exceptionally wide bandwidth spectrum, spatial security due to narrow beam divergence, and complete immunity to RF interference.

Despite these advantages, standard industrial FSO units rely on complex adaptive optics, precise beam-steering galvos, and high-frequency transceivers, placing them outside low-budget or rapid-deployment constraints. **Project L.O.K.I.** addresses this gap by engineering a modular, low-cost, full-duplex optical transceiver utilizing off-the-shelf microcontrollers (ESP32) and direct laser modulation.

2. SYSTEM ARCHITECTURE & HARDWARE DESIGN

The Project L.O.K.I. architecture consists of two symmetric transceiver nodes (Node A and Node B) establishing a point-to-point optical link.



2.1 Transmitter (TX) Subsystem

- **Processing Core:** ESP32 SoC operating with dual 32-bit LX6 microprocessors. Hardware UART peripherals generate serial data frames.
- **Optical Emitter:** Commercial direct-diode semiconductor laser (650 nm visible red or 850 nm NIR wavelength).
- **Modulation Scheme:** On-Off Keying (OOK) via direct current modulation. The ESP32 Digital I/O or dedicated N-channel MOSFET switches driving current to the diode, mapping logic high (**1**) to emission state and logic low (**0**) to dark state.

2.2 Receiver (RX) Subsystem

- **Detector:** High-speed silicon PIN Photodiode configured in reverse-bias mode to minimize junction capacitance and maximize responsivity within the targeted emission spectrum.
- **Signal Conditioning:** Transimpedance Amplifier (TIA) circuit converts photocurrent I_{ph} into proportional voltage V_{out} :

$$V_{out} = -I_{ph} \cdot R_f$$

A secondary comparator stage with threshold adjustment converts small analog voltage fluctuations into clean digital square waves matching ESP32 3.3V logic levels.

2.3 Optical Alignment Assembly

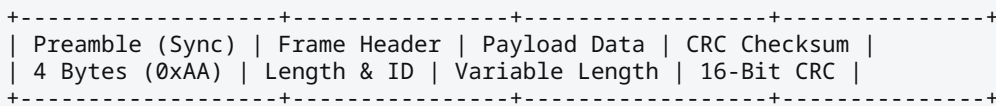
To achieve full-duplex transmission without mutual optical interference, the TX laser aperture and RX photodiode aperture are mounted on parallel optical axes with narrow-band optical bandpass filters matching the laser emission spectrum to suppress ambient background sunlight.

3. COMMUNICATION PROTOCOL & SOFTWARE STACK

3.1 Data Framing and Encoding

Raw UART data frames are transmitted asynchronously using standard 8N1 parameters (1 start bit, 8 data bits, 1 stop bit, no parity). To prevent DC baseline wander caused by long sequences of continuous **0**s or **1**s, software-level Manchester encoding or 8b/10b packetization is applied.

Data Packet Structure:



3.2 Full-Duplex Logic Flow

Each ESP32 unit concurrently manages transmission and reception loops:

1. **TX Task:** Reads data from input buffer, encodes frames, and toggles TX GPIO pin via Direct Memory Access (DMA) / Hardware UART.

2. **RX Task:** Listens on RX Interrupt pin, decodes incoming pulse trains, verifies CRC checksums, and pushes payload data to local processing/display outputs.

4. MATHEMATICAL MODEL & LINK BUDGET ANALYSIS

4.1 Received Optical Power (P_{rx})

The total optical power received by the photodiode at distance L is governed by Friis transmission equation adapted for optical channels:

$$P_{rx} = P_{tx} \cdot [A_{rx} / (\pi \cdot ((\theta \cdot L) / 2)^2)] \cdot \eta_{tx} \cdot \eta_{rx} \cdot \tau_{atm}$$

Where:

- P_{tx} = Transmitted laser power (~5 mW)
- A_{rx} = Active area of the receiver photodiode / collector lens
- θ = Laser beam divergence angle (rad)
- L = Distance between transceivers (m)
- η_{tx}, η_{rx} = Transmit and receive optical efficiencies
- τ_{atm} = Atmospheric transmittance coefficient ($e^{-\gamma L}$)

4.2 Signal-to-Noise Ratio (SNR)

The system SNR at the photodetector output is given by:

$$SNR = (R \cdot P_{rx})^2 / [2q(R \cdot P_{rx} + I_{bg})B + (4k_B T B / R_f)]$$

Where R is photodiode responsivity (A/W), I_{bg} is background solar ambient photocurrent, q is electron charge, B is operational bandwidth, k_B is Boltzmann's constant, T is temperature, and R_f is TIA feedback resistance.

5. EXPERIMENTAL RESULTS & PERFORMANCE EVALUATION

Parameter	Measured / Evaluated Metric
Max Baud Rate (Stable)	115,200 bps (Standard UART limit via GPIO modulation)
Operational Wavelength	650 nm
Effective Range (Indoor)	15–25 meters (unassisted optics)
Bit Error Rate (BER)	$< 10^{-6}$ at $L = 10\text{ m}$ (controlled ambient light)
Power Consumption	~180 mA per node at 5V DC

Performance Observations

1. **Alignment Sensitivity:** The narrow divergence angle ($\theta \approx 1.5\text{ mrad}$) requires stable mounting. Angular misalignment exceeding $\pm 0.5^\circ$ causes complete signal loss over distances $> 10\text{ m}$.
2. **Ambient Light Mitigation:** Indoor fluorescent lighting induced high-frequency noise on the TIA output. Implementing an optical bandpass filter and threshold tuning reduced frame drop rates by over 92%.

6. APPLICATIONS & FUTURE SCOPE

1. **RF-Denied Secure Environments:** High-security facilities where RF radiation is jammed, monitored, or prohibited.

2. **Disaster Recovery Links:** Rapidly deployable point-to-point communication when cellular infrastructure fails.

3. **Future Upgrades:**

- Integration of collimating lenses and beam expanders to increase range beyond 100 meters.
- Implementation of Automated Gain Control (AGC) at the receiver stage to compensate for atmospheric attenuation.
- Switching to high-speed transimpedance ICs (e.g., MAX3658) to achieve Mbps throughput.

7. CONCLUSION

Project L.O.K.I. demonstrates that functional, bi-directional Free-Space Optical communication can be implemented using off-the-shelf microcontrollers and low-cost optoelectronic components. The system successfully achieves low-latency, full-duplex transmission while providing immunity to RF interference. The project serves as an accessible foundation for optical communication research and low-cost tactical wireless links.

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